

Laird Scabar

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EDUCATION

University of British Columbia

Bachelor of Applied Science — Mechanical Engineering, Mechatronics

Vancouver, BC

GPA: 4.20/4.33

SKILLS

Design & Analysis: CAD (Catia, SolidWorks, Revit), FEA (Ansys), DFM/DFA, GD&T, Tolerance Analysis, PCB

Manufacturing & Prototyping: 3D Printing (FDM/SLA), CNC, Manual Mill/Lathe, Welding (Stick), Water Jet

Testing & Validation: MATLAB, Experiment Design, Failure Analysis, C, Circuit Design

Certifications: Certified Solidworks Design Associate (CSWA)

EXPERIENCE

Mechanical Engineering Intern – Thermal/HVAC

May 2026 – Sept 2026

VETS Group

Burnaby, BC

- Designed custom sheet metal components in SolidWorks for client-specific HVAC and dust collection applications, producing manufacturable models and drawings that were pushed to shop production.
- Created HVAC system layouts in Revit by routing ductwork, placing mechanical equipment, and coordinating system geometry to support project design and installation planning.
- Collaborated with engineers, project managers, and fabrication teams to review design requirements, resolve fit-up constraints, and ensure sheet metal parts met client specifications before manufacturing.

SUBC - UBC's Submarine Design Team

Sept 2025 – Present

Drive Train Sub-team Member

Vancouver, BC

- Designed drivetrain components in SolidWorks using DFM principles, producing manufacturable CAD models and fabrication drawings for gearbox subsystem integration.
- Performed FEA stress and deformation analysis to improve structural reliability, reducing peak stress concentrations by ~30% through iterative design changes.
- Prepared CNC-ready components with appropriate tolerancing and supported cross-team drivetrain integration, improving assembly accuracy, maintenance access, and overall build efficiency.

PROJECTS

Gearbox Side Plate & Step-Down Collar | *SolidWorks, FEA, CNC, DFM*

October 2025

- Designed, optimized, and fabricated gearbox side plate using CAD and FEA insights, reducing weight by 45% while maintaining required safety factors under peak torque loads.
- Conducted stress and deformation analysis to guide material removal, improving strength-to-weight ratio.
- Contributed to the design and fabrication of a custom step-down collar for our smaller output gear through CAD iteration and fit checks, eliminating slippage and restoring drivetrain alignment.

Autonomous Robotic Claw | *C++, SolidWorks (Sheet metal), Rapid Prototyping*

January 2026

- Led the design and fabrication of an automated robotic claw capable of detecting and grasping objects using an ultrasonic sensor and servo-driven mechanism.
- Programmed Arduino micro-controller in C++ to implement distance-based actuation logic, enabling autonomous object pickup without manual control.
- Integrated mechanical, electrical, and software subsystems while troubleshooting sensor noise and servo calibration, improving detection reliability to more than 95% during testing.

Self-Balancing Robot | *C++, Arduino, MPU-6050, Control Systems*

December 2025

- Built a two-wheel self-balancing robot using an MPU-6050 IMU for real-time tilt estimation and sensor fusion.
- Implemented a PID control algorithm to maintain dynamic stability under external disturbance.
- Developed closed-loop motor control using an L298N driver to regulate dual DC motors based on feedback signals.

ADDITIONAL EXPERIENCE

Film & Television Actor

2015–2025

- Performed in professional productions, developing strong communication, adaptability, and teamwork skills.
- Worked with companies including Netflix, Disney, CBC, Pfizer, and more.

Terry Fox Foundation — Event Coordinator

Sept 2020 – Present

- Supported fundraising events and logistics, contributing to community engagement and event execution.
- Contributed to raising over \$50,000 for cancer research.